Subtractors -

The subtraction of 2 binary nos. may be accomplished by taking the complement of subtrahend & adding it to the minuend.

$$\begin{array}{l} A \text{ (Minuend)} \\ \underline{-B \text{ (subtrahend)}} \end{array} \Rightarrow \begin{array}{l} A \\ \underline{+ \text{ 2's complement of } B} \end{array} \Rightarrow$$

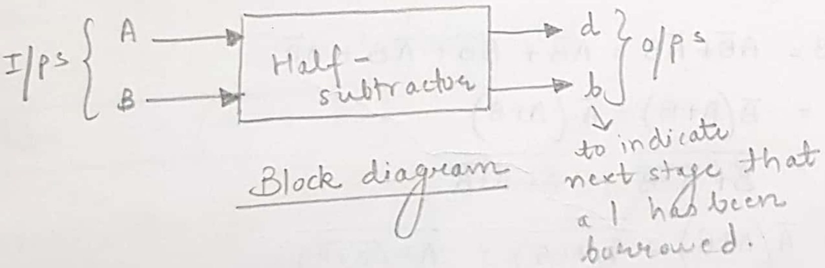
sub.ⁿ becomes an addition operation.

∴, Instead of having separate circuit for subⁿ, adder itself can be used to perform sub.ⁿ. This results in red.ⁿ of h/w.

* If minuend bit is smaller than subtrahend $(A<B)$, a 1 is borrowed from the next significant position. This must be conveyed to the next higher pair of bits by means of a signal coming out of a given stage (bout) & going as i/p to the next higher stage.

The Half - Subtractor -

A half-subtractor is a combinational circuit that subtracts one bit from the other & produces the difference. It also has an o/p to specify if a 1 has been borrowed.



Block diagram

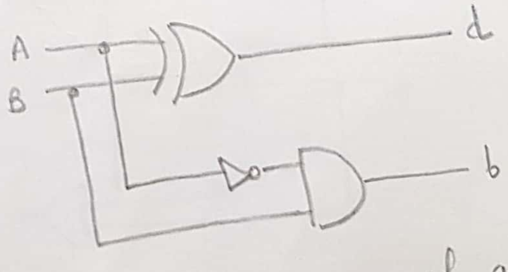
Inputs		Outputs	
A	B	d	b
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Truth table

$d = \bar{A}B + A\bar{B}$

$ = A \oplus B$

$b = \bar{A}B$ } by ANDing the complement of minuend with the subtrahend.



Logic diagram of a half - subtractor

Realisation of a half-subtractor using universal gates -

NAND Logic -

$$d = A \oplus B = \bar{A}B + A\bar{B}$$

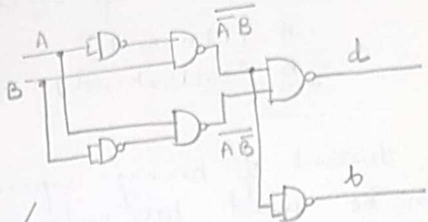
$$\bar{d} = \overline{\bar{A}B + A\bar{B}}$$

$$d = \bar{\bar{d}} = \overline{\overline{\bar{A}B}\cdot\overline{A\bar{B}}}$$

$$b = \bar{A}.B$$

$$\bar{b} = \overline{\bar{A}.B}$$

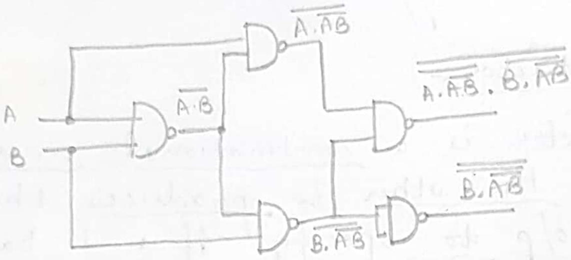
$$b = \bar{\bar{b}} = \overline{\overline{\bar{A}.B}}$$



This is not minimal.

$$b = \bar{A}B = B(\bar{A}+\bar{B}) = B(\overline{AB}) = \overline{\overline{B.\overline{AB}}}$$

$$d = A \oplus B = \bar{A}B + A\bar{B} = \overline{\overline{B.\overline{AB}}\cdot\overline{A.\overline{AB}}}$$



HS using only 2-input NAND gates

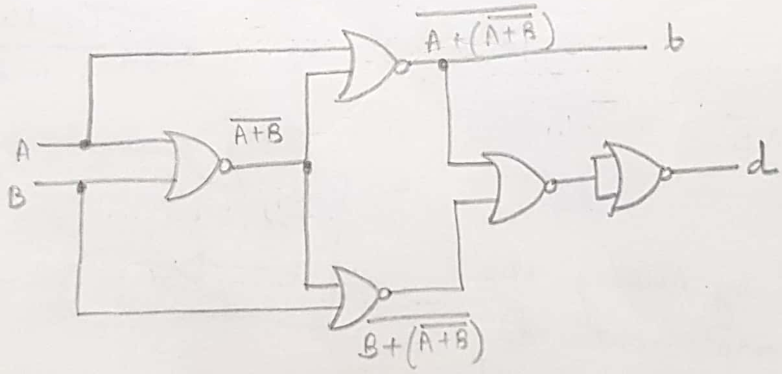
NOR Logic -

$$d = A \oplus B = A\bar{B} + \bar{A}B = A\bar{B} + B\bar{B} + \bar{A}B + A\bar{A}$$

$$= \bar{B}(A+B) + \bar{A}(A+B)$$

$$= \overline{B + \overline{A+B}} + \overline{A + \overline{A+B}}$$

$$b = \bar{A}B = \bar{A}(A+B) = \overline{\overline{\bar{A}(A+B)}} = \overline{A + \overline{(A+B)}}$$

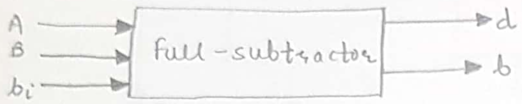


HS using only 2-input NOR gates

The full - Subtractor -

A full - Subtractor subtracts one bit (B) from another bit (A), when already there is a borrow b_i from this column for the subtraction in the preceding column & outputs the difference bit (d) & the borrow bit (b) required from the next column.

A FS is a combinational circuit with <u>three inputs (A, B, b_i)</u> and <u>2 o/ps d and b</u>.



Block diagram

Inputs A	B	b_i	Difference d	Borrow b
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Truth Table

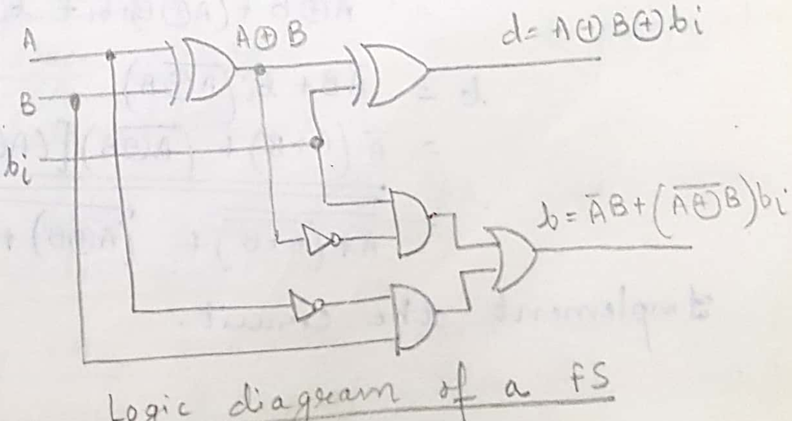
$$d = \bar{A}\bar{B}b_i + \bar{A}B\bar{b_i} + A\bar{B}\bar{b_i} + ABb_i$$
$$= b_i(AB + \bar{A}\bar{B}) + \bar{b_i}(\bar{A}B + A\bar{B})$$
$$= b_i(\overline{A \oplus B}) + \bar{b_i}(A \oplus B)$$
$$\boxed{d = A \oplus B \oplus b_i}$$

$$b = \bar{A}\bar{B}b_i + \bar{A}B\bar{b_i} + \bar{A}Bb_i + ABb_i$$
$$= \bar{A}B(b_i + \bar{b_i}) + b_i(AB + \bar{A}\bar{B})$$
$$\boxed{b = \bar{A}B + (\overline{A \oplus B})\, b_i}$$

Also, $b = \bar{A}\bar{B}b_i + \bar{A}B\bar{b_i} + \bar{A}Bb_i + ABb_i$
$$= \bar{A}\bar{B}b_i + \bar{A}B\bar{b_i} + ABb_i + \bar{A}Bb_i + \bar{A}Bb_i + \bar{A}Bb_i$$
$$= \bar{A}b_i(B + \bar{B}) + \bar{A}B(b_i + \bar{b_i}) + Bb_i(A + \bar{A})$$
$$\boxed{b = \bar{A}b_i + \bar{A}B + Bb_i}$$

A \ Bb_i	$\bar{B}\bar{b_i}$ 00	$\bar{B}b_i$ 01	Bb_i 11	$B\bar{b_i}$ 10
$\bar{A}$ 0		1	1	1
A 1			1	

$b = \bar{A}b_i + \bar{A}B + Bb_i$



Logic diagram of a FS